Database Design



Why? Benefits of good design

- Simplicity
- Speed
- Integrity
- Adaptability



How? Data modeling

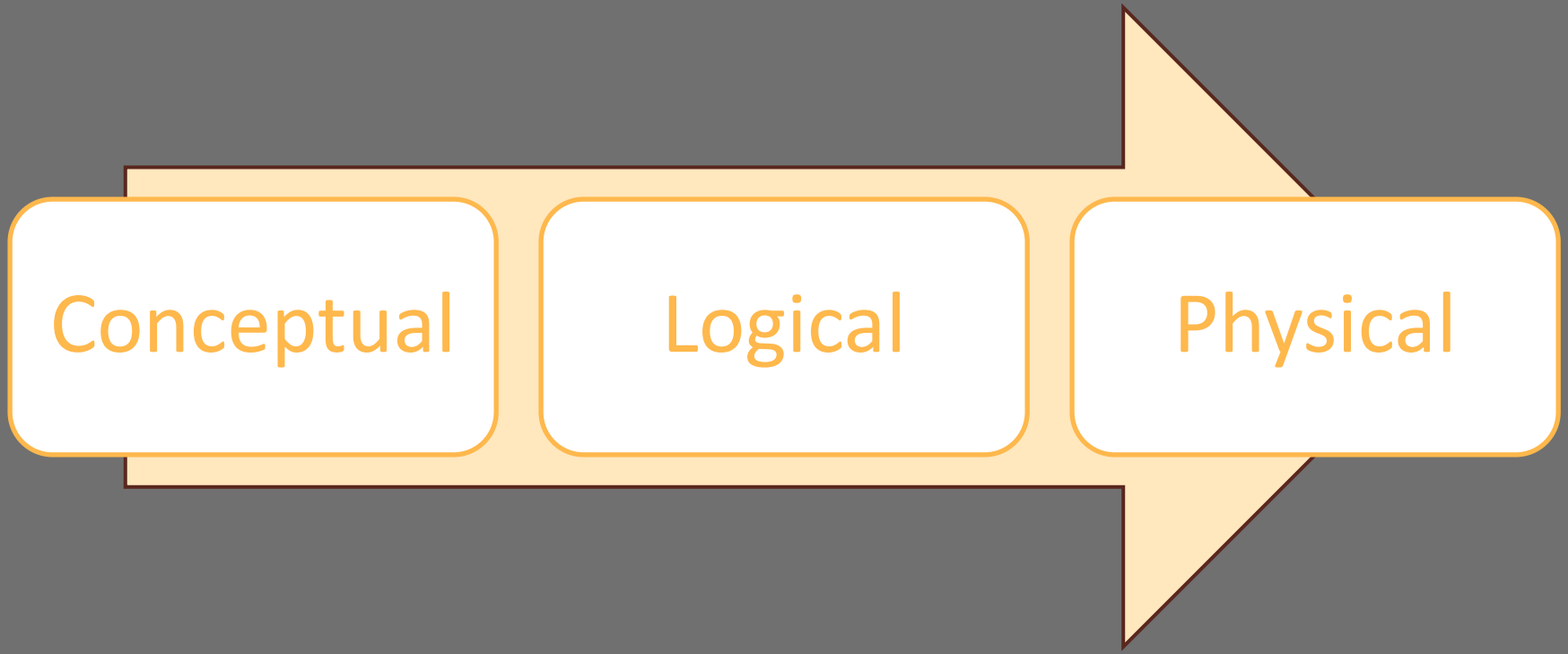
“a notation for describing data or information”

- Garcia-Molina, Ullman, Widom

“a collection of conceptual tools for describing data, data relationships, data semantics and consistency constraints”

- Silbershatz, Korth, Sudarshan

Data modeling: 3 types/phases



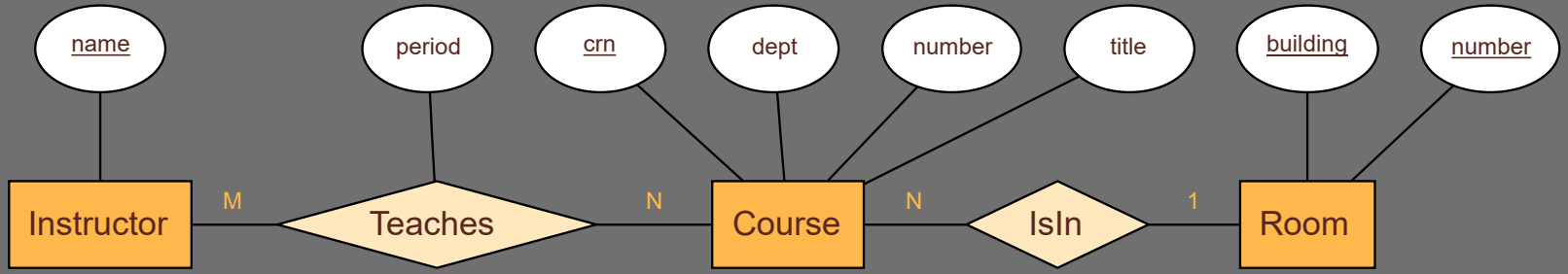
Notation: Entity-Relationship Diagrams



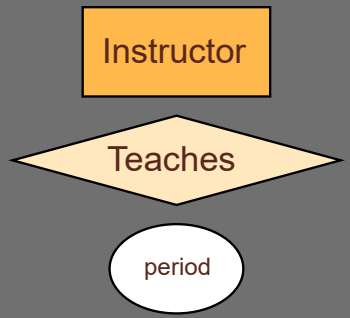
“The Entity-Relationship model – toward a unified view of data”

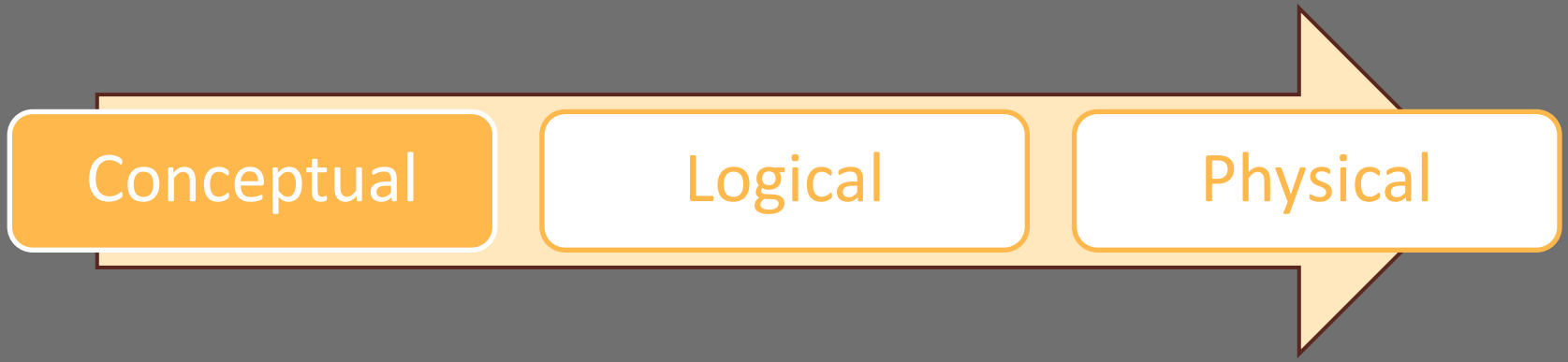
Peter Chen, 1976

Notation: Entity-Relationship Diagrams



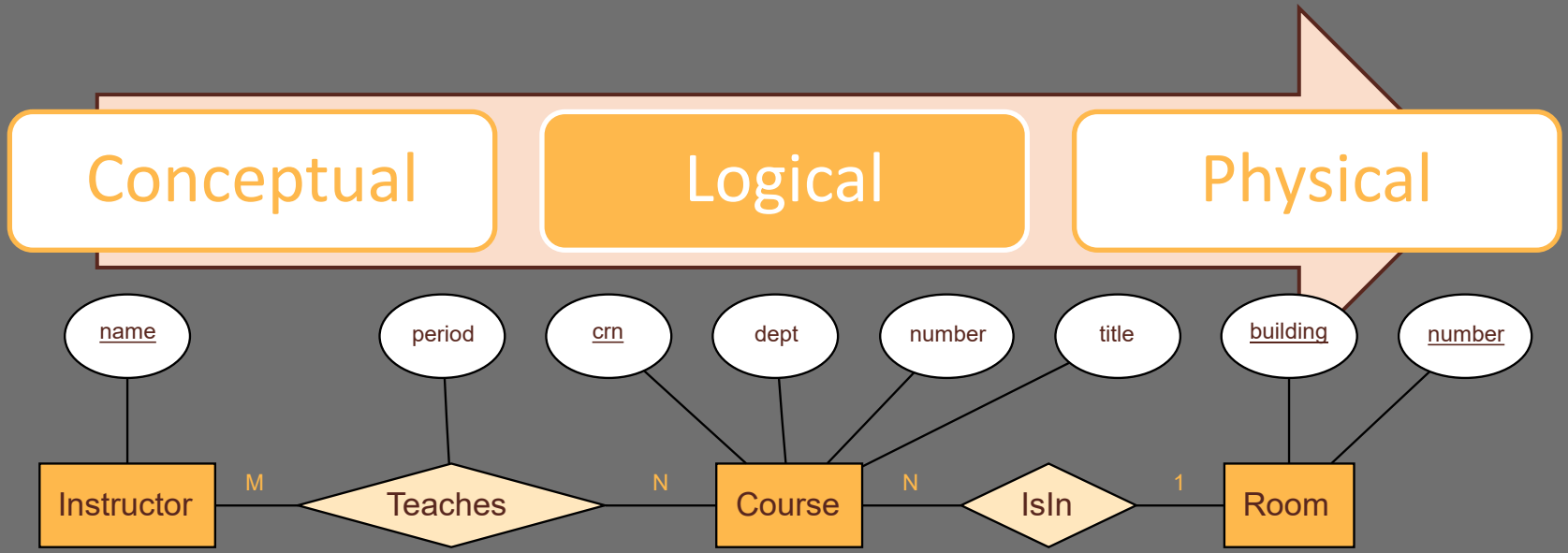
- Entity sets
- Relationships
- Attributes



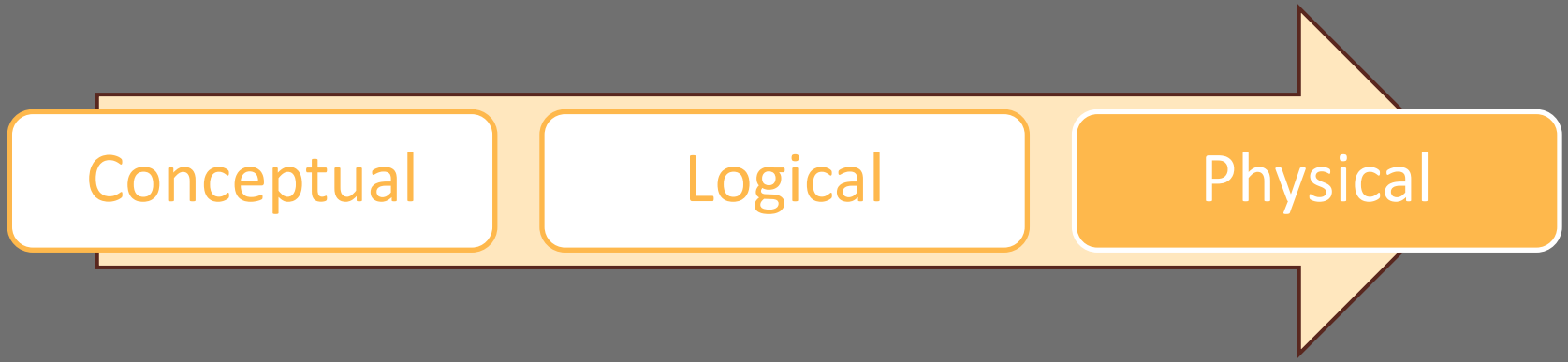




Decision: Entity set vs. relationship (vs. attribute)



Decision: Semantically-equivalent ER models



Instructor
name: VARCHAR (PK)

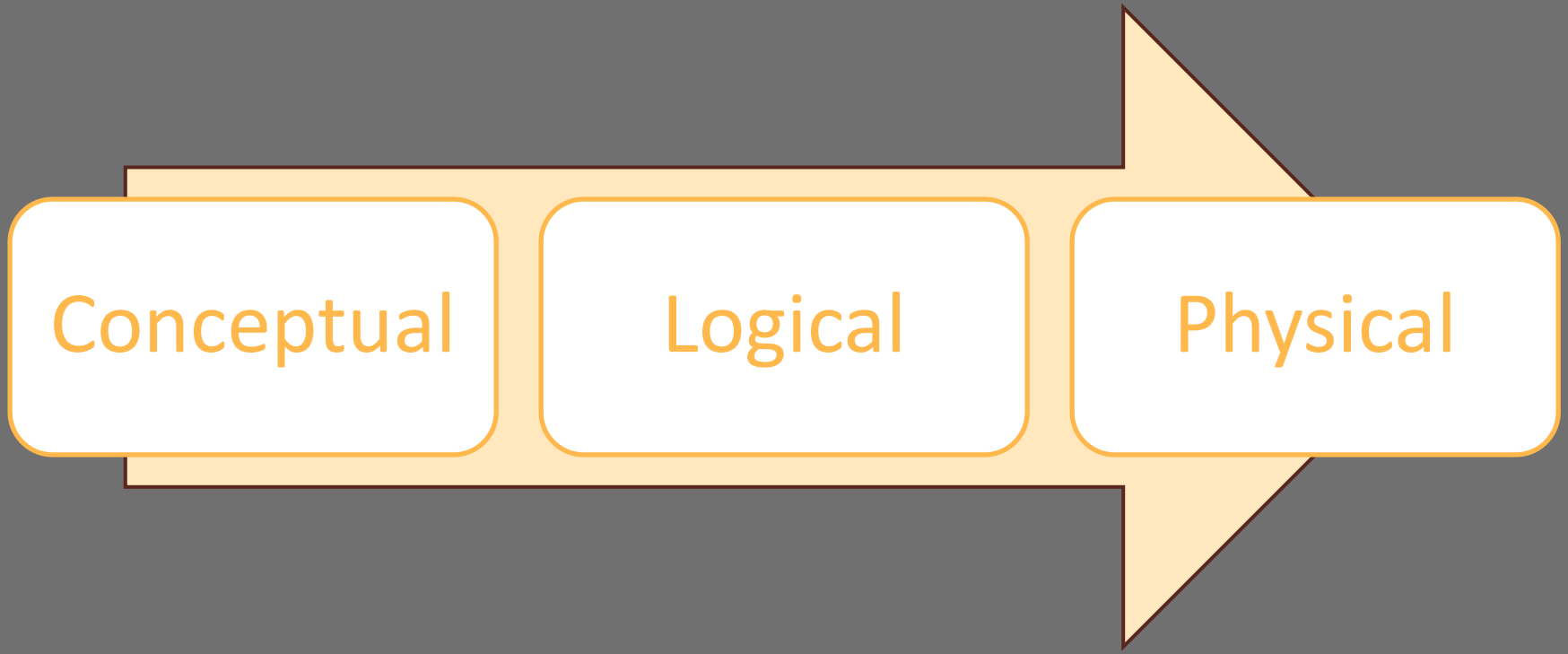
Room
building: VARCHAR (PK)
number: INT (PK)

Teaches
name: VARCHAR (PK)
crn: VARCHAR (PK)
period: VARCHAR

Course
crn: VARCHAR (PK)
dept: CHAR(4)
number: CHAR(3)
title: VARCHAR
building: VARCHAR (FK)
room: INT (FK)

Decision: Near-equivalent SQL implementations of ER model

Continuum



Comments about ERD notation

- Numerous variations
- Similar to Unified Modeling Language (UML)



Why not model in SQL?

- Appropriate for Physical model.

- Not appropriate for non-technical discussions.
- Requires implementation decisions.
- SQL can change.

How? Normalization

- Objective definition of good design
- Algorithm

Some problems of poor design

Mistake	Can lead to …
Redundant attributes	Same data multiple times; Inconsistent data
Unrelated attributes in same entity set	Redundancy
All attributes on just one entity set	Redundancy
Related attributes in wrong entity sets	More joins
Entity sets without any relationships	No referential integrity; Difficulty joining
Too many key attributes	Slower lookup; Data unnecessarily constrained



Other database design issues

Later:

- How DB interfaces with applications
- Data validation
- Security
- Concurrency
- Fault-tolerance
- …